

Worksheet 1

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Subject Name: Internet Of Things

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1. Aim/Overview of the practical:

Interface an Arduino Uno with LEDs to design the different patterns and follow the different cases. Use 6 different color LEDs and name them T1 to T6 and use resistors of suitable values.

- T1, T2 and T3 are in ON condition and T4, T5 and T6 are in Off conditions.
- T1, T2 and T3 are in OFF condition and T4, T5 and T6 are in ON conditions.
- T1, T3 and T5 are in ON conditions and T2, T4 and T6 are in OFF conditions.
- T1, T3 and T5 are in OFF conditions and T2, T4 and T6 are in ON conditions.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno Board
- Breadboard
- LEDs (Light Emitting Diodes)
- 220 Ω Resistors
- Jumper Wires
- Computer/Laptop
- 5V Power Supply (USB or Adapter)
- USB Cable (Type-A to Type-B)

Software requirements

- Arduino IDE
- Arduino Uno Board Drivers
- Windows 10/11, Linux, or macOS

Coding/Circuit Diagram:

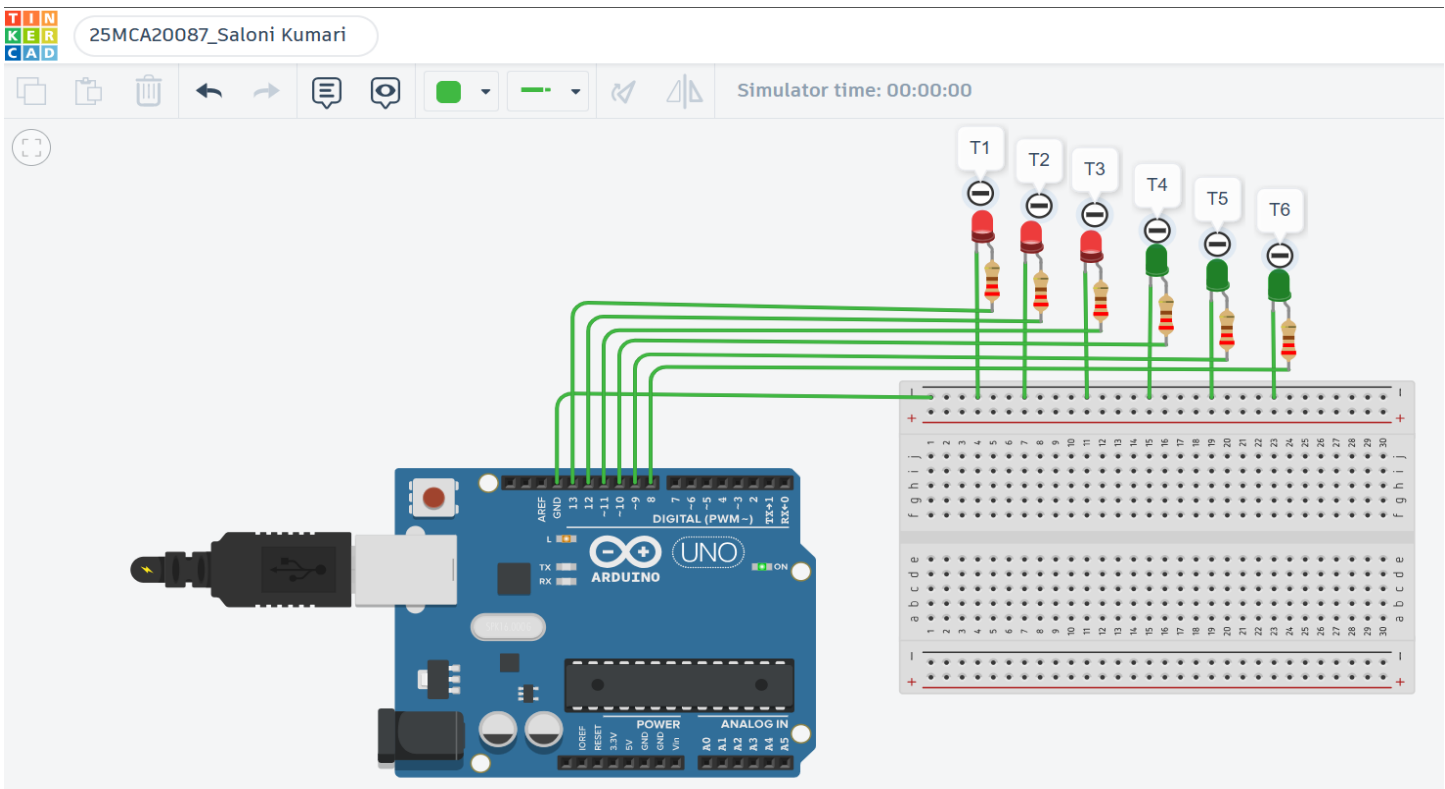
```
// Assign Arduino pin numbers to LEDs  
int led1 = 13;  
int led2 = 12;  
int led3 = 11;  
int led4 = 10;  
int led5 = 9;  
int led6 = 8;
```

Case1

```
void setup()
{
    // Set all LED pins as OUTPUT
    // This tells Arduino that these pins will send signals to LEDs.
    pinMode(led1, OUTPUT);
    pinMode(led2, OUTPUT);
    pinMode(led3, OUTPUT);
    pinMode(led4, OUTPUT);
    pinMode(led5, OUTPUT);
    pinMode(led6, OUTPUT);
}

void loop()
{
    // Pattern 1
    // Turn ON LED1, LED2, LED3
    // Turn OFF LED4, LED5, LED6
    digitalWrite(led1, HIGH); // LED1 ON
    digitalWrite(led2, HIGH); // LED2 ON
    digitalWrite(led3, HIGH); // LED3 ON
    digitalWrite(led4, LOW); // LED4 OFF
    digitalWrite(led5, LOW); // LED5 OFF
    digitalWrite(led6, LOW); // LED6 OFF

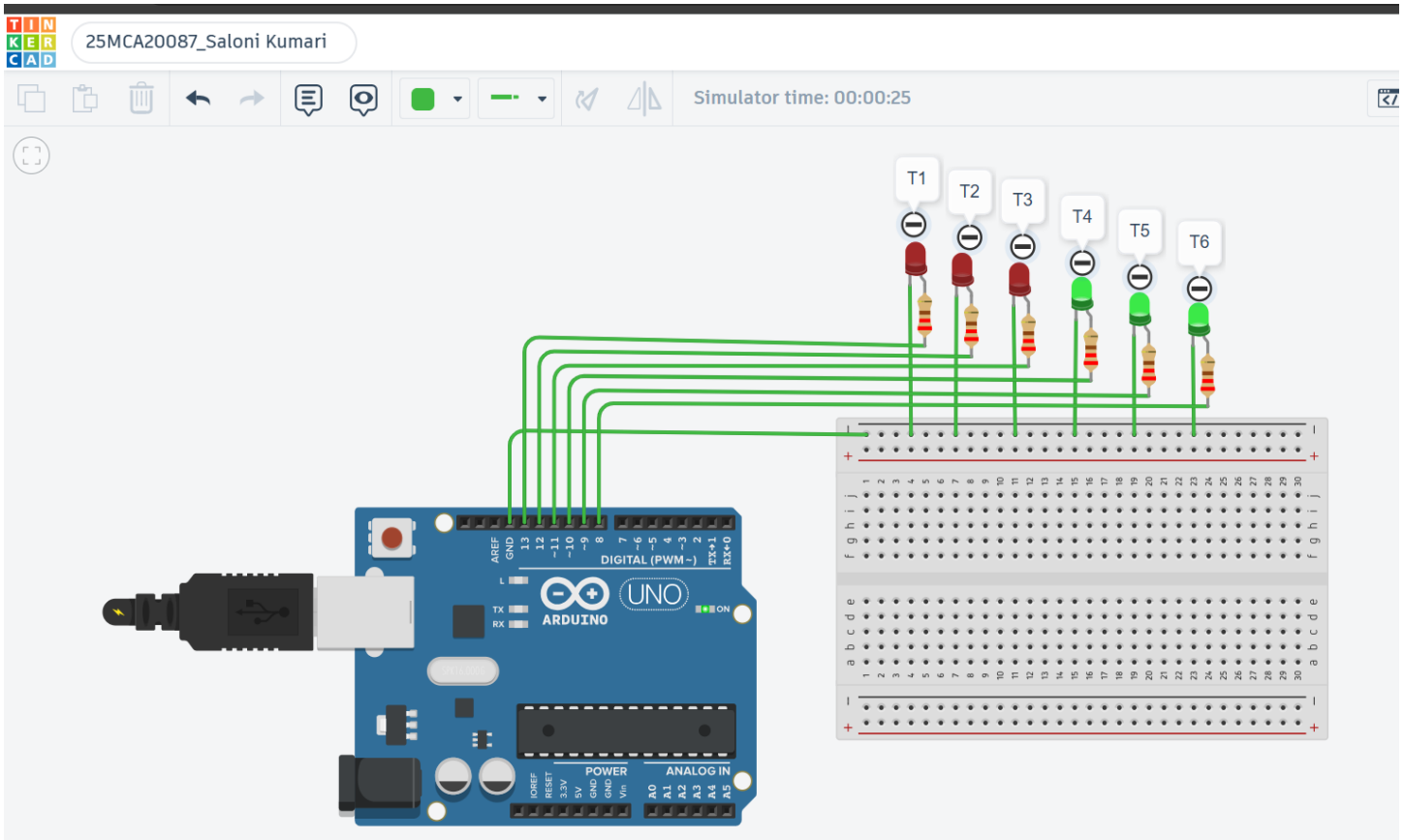
    delay(1000); // Wait for 1 second
```



Case2

```
// Turn ON LED4, LED5, LED6
// Turn OFF LED1, LED2, LED3
digitalWrite(led1, LOW); // LED1 OFF
digitalWrite(led2, LOW); // LED2 OFF
digitalWrite(led3, LOW); // LED3 OFF
digitalWrite(led4, HIGH); // LED4 ON
digitalWrite(led5, HIGH); // LED5 ON
digitalWrite(led6, HIGH); // LED6 ON

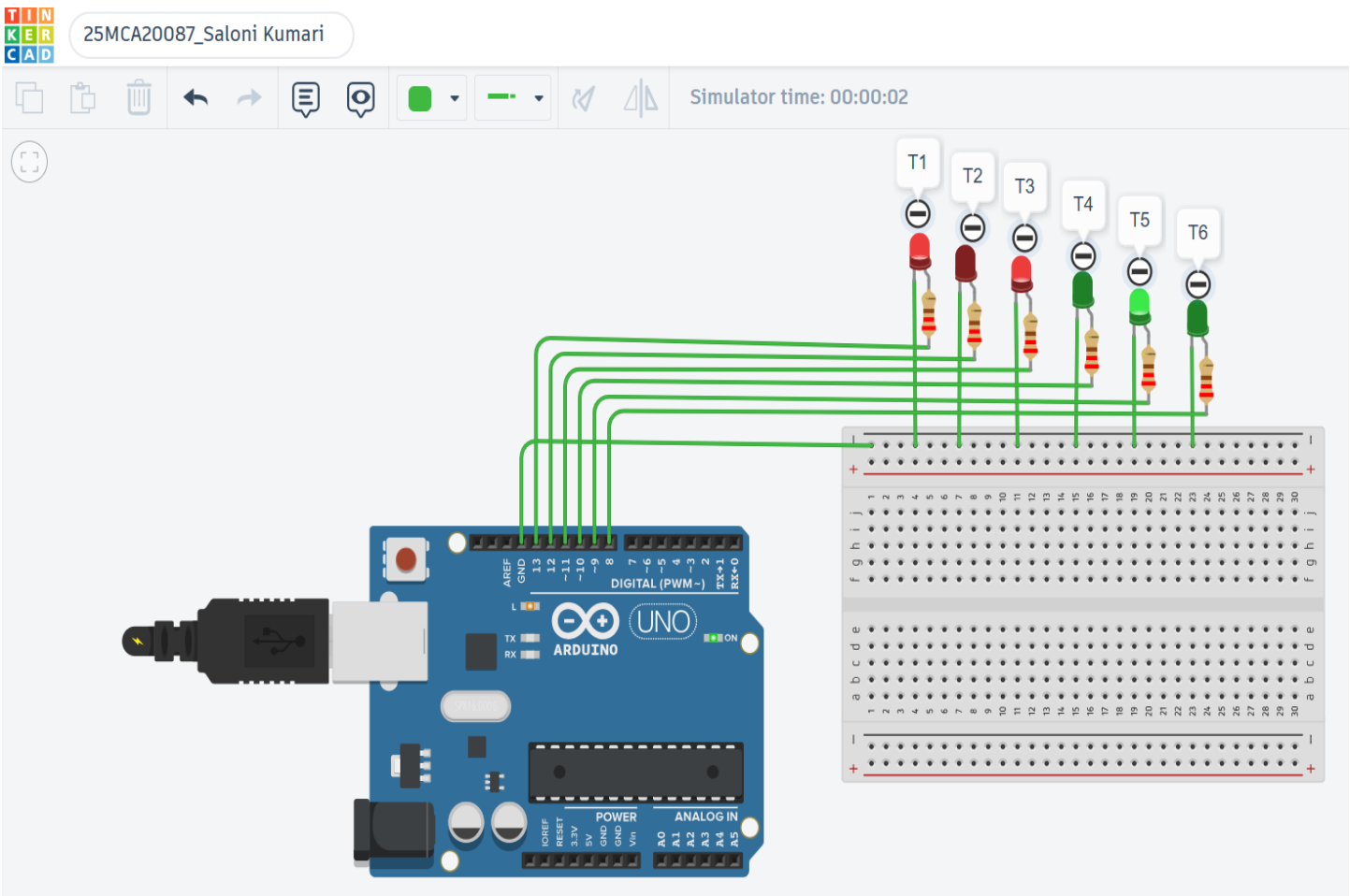
delay(1000); // Wait for 1 second
```



Case3

```
// Turn ON odd LEDs (1,3,5)
// Turn OFF even LEDs (2,4,6)
digitalWrite(led1, HIGH); // LED1 ON
digitalWrite(led2, LOW);  // LED2 OFF
digitalWrite(led3, HIGH); // LED3 ON
digitalWrite(led4, LOW);  // LED4 OFF
digitalWrite(led5, HIGH); // LED5 ON
digitalWrite(led6, LOW);  // LED6 OFF
```

```
delay(1000); // Wait for 1 second
```

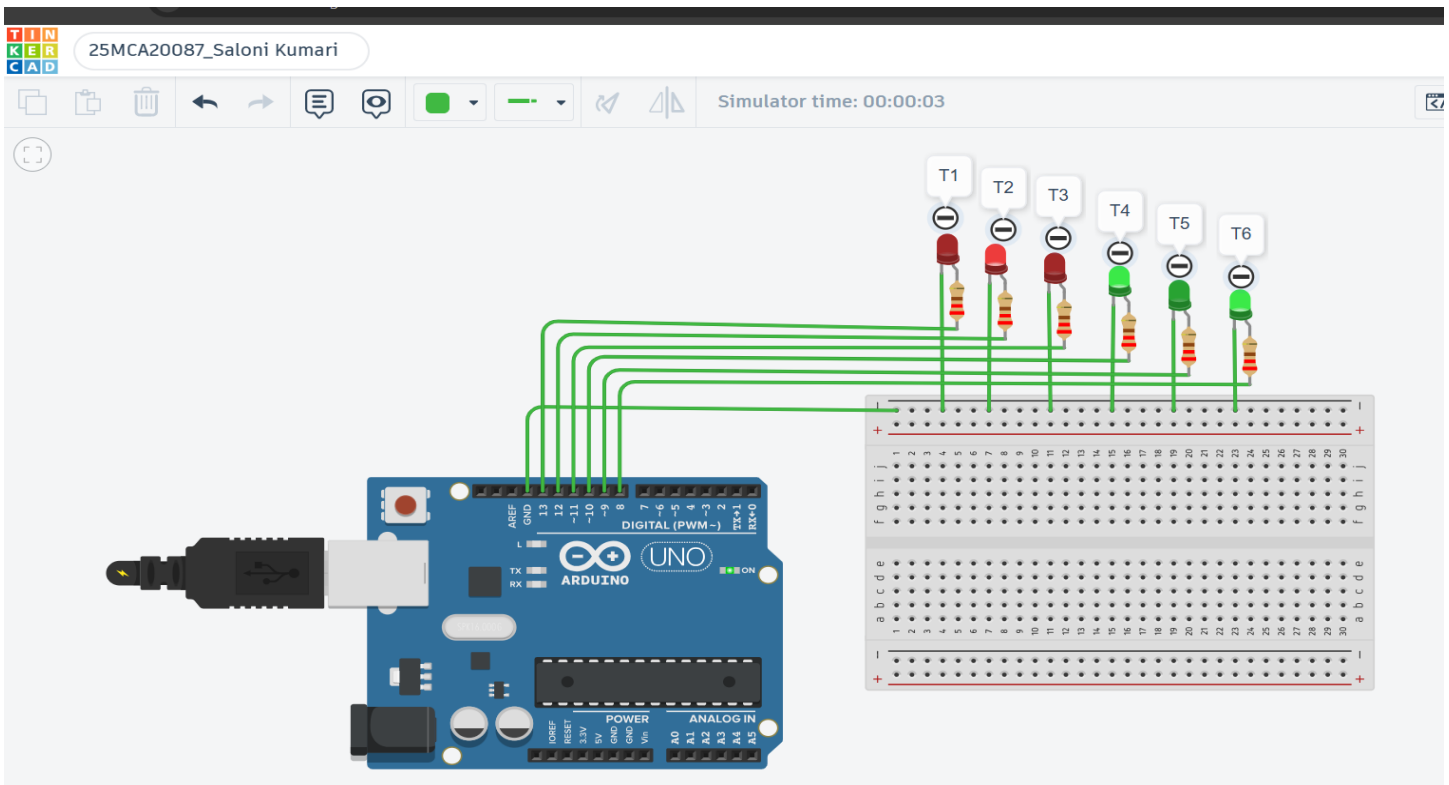


Case4

```
// Turn ON even LEDs (2,4,6)
// Turn OFF odd LEDs (1,3,5)
digitalWrite(led1, LOW); // LED1 OFF
digitalWrite(led2, HIGH); // LED2 ON
digitalWrite(led3, LOW); // LED3 OFF
digitalWrite(led4, HIGH); // LED4 ON
digitalWrite(led5, LOW); // LED5 OFF
digitalWrite(led6, HIGH); // LED6 ON

delay(1000); // Wait for 1 second

// After this, loop() starts again automatically,
// so the LED patterns repeat continuously.
}
```



Learning outcomes (What I have learnt):

- Understand the basic working of the Arduino Uno board.
- Learn how to connect LEDs with Arduino using a breadboard.
- Write and upload Arduino programs using the Arduino IDE.
- Use `pinMode()` to configure digital pins as outputs.
- Control LEDs using `digitalWrite()` with HIGH and LOW signals.
- Use the `delay()` function to create LED blinking and timing effects.
- Design and implement different LED lighting patterns.